Online Resource 1: Detailed ablations and robustness analyses for calibration-free two-view 3D pose fusion

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1 Purpose

This Online Resource reports prespecified and post hoc ablations, synthetic corruption settings, robustness diagnostics and secondary cohort results for the accompanying article. It preserves the separation between training evidence and evaluation evidence: triangulated poses were resolved only by the evaluation layer and were never used for model training, validation or checkpoint selection.

2 Ablation definitions

3 Complete held-out learned comparison

All A2–A9 outputs were view-order invariant to the reported precision. A3-relative (ROM, peak-speed) retention was A0 (1.156, 1.018), A1 (1.050, 1.654), A2 (1.101, 0.917), A3 (1.000, 1.000), A4 (0.949, 0.828), A5 (0.943, 0.644), A6 (1.000, 1.000), A7 (0.948, 0.821), A8 (1.218, 2.187) and A9 (0.997, 2.615). The A8 and A9 variants demonstrate that relaxing the conservative correction can increase coordinate error and over-amplify peak angular speed.

Table 1 Ablation design. A4–A9 share the same network. Checkmarks indicate active loss groups. A7–A9 are post hoc trunk-twist variants and do not replace the prespecified complete model A6.

ID	Method	Active components
A0	Face stream only	Deterministic single-view output
A1	Side stream in face reference	Deterministic single-view output
A2	Canonical arithmetic mean	Deterministic mean
A3	Deterministic quality mean	Fixed quality weighting
A4	Spatial-objective fusion	Learned residual; recovery, identity and skeletal losses
A5	Rotation-temporal fusion	A4 plus rotation and temporal losses
A6	Complete self-supervised fusion	A5 plus complete-cycle ROM loss
A7	A6 + per-view-peak ROM anchor	A6 plus post hoc trunk-twist anchor
A8	A7 + rotation-parameterized residual	A7 plus post hoc twist residual
A9	A8 + observed twist-rate anchor	A8 plus post hoc rate anchor

Table 2 Held-out comparison ($N = 14$). Mean per-joint position error is reported after one similarity alignment per cycle followed by framewise hip centring to the same-video triangulated pseudo-reference. Differences are method minus A6 with participant-bootstrap 95% confidence intervals and Holm-adjusted paired Wilcoxon p values. Rotation range of motion (ROM) and peak angular speed are A3-relative retention ratios, not comparisons with independent kinematics.

ID	Method	Error (mm)	Difference [95% CI]	p_{Holm}
A0	Face only	77.92 ± 12.52	+17.14 [13.69, 21.34]	0.0011
A1	Side only	75.84 ± 13.93	+15.07 [10.23, 19.48]	0.0018
A2	Arithmetic mean	60.85 ± 8.98	+0.07 [−0.50, 0.59]	1.0000
A3	Quality mean	61.03 ± 8.84	+0.26 [0.17, 0.35]	0.0011
A4	Spatial objectives	62.81 ± 8.86	+2.04 [1.71, 2.36]	0.0011
A5	Rotation/temporal	60.80 ± 8.96	+0.02 [−0.38, 0.43]	1.0000
A6	Complete model	60.78 ± 8.72	–	–
A7	Per-view-peak ROM	61.31 ± 8.87	+0.54 [0.38, 0.69]	0.0011
A8	Twist residual	92.02 ± 9.59	+31.25 [27.78, 34.88]	0.0011
A9	Twist-rate anchor	94.11 ± 9.14	+33.34 [30.00, 36.77]	0.0011

4 Deterministic comparison

Table 3 Agreement with the private triangulated pseudo-reference over 137 participants. Values are mm after one similarity alignment per cycle followed by framewise hip centring. The pseudo-reference-fitted row is a leakage diagnostic and is not an eligible label-free method.

Method	Mean	SD	95% CI
Body-frame average	65.249	14.523	[62.935, 67.789]
World-coordinate average	123.703	18.096	[120.710, 126.803]
Root alignment and average	123.709	18.096	[120.678, 126.787]
Similarity alignment, all joints	69.317	18.250	[66.358, 72.465]
Similarity alignment, stable joints	66.086	14.291	[63.719, 68.557]
Pseudo-reference-fitted joint weights (leaky)	64.077	14.449	[61.718, 66.539]
Similarity alignment, body-part weights	66.270	14.593	[63.863, 68.745]
Similarity alignment, side smoothing	67.037	14.268	[64.686, 69.437]
Similarity alignment, output smoothing	68.031	14.298	[65.731, 70.435]

In this secondary all-participant analysis, Extrinsic-R reduced the mean by 2.175 mm relative to deterministic body-frame averaging and was lower for 109 of 137 participants ($p_{\text{Holm}} = 1.24 \times 10^{-14}$). The quality-weighted variant reduced the mean by 1.116 mm and was lower for 89 participants ($p_{\text{Holm}} = 4.92 \times 10^{-5}$). Hold-out reprojection error was associated with Extrinsic-R error (Spearman $\rho = 0.347$, $p = 3.35 \times 10^{-5}$). These results describe all available participants, including training and validation participants, and are not an independent learned-model test.

Table 4 Secondary camera-assisted deterministic comparison over all 137 participants. The difference is method minus the calibration-free body-frame average, with a participant-bootstrap 95% confidence interval. p values are Holm-adjusted paired Wilcoxon tests. Each cycle uses one similarity alignment to the same-video pseudo-reference followed by framewise hip centring.

Method	MPJPE (mm)	Difference [95% CI]	p_{Holm}	Improved people
Body-frame average	65.249 ± 14.523	–	–	–
Extrinsic-R	63.074 ± 14.962	-2.175 [-2.622, -1.730]	1.24e-14	109/137
Extrinsic-R quality	64.133 ± 15.301	-1.116 [-1.613, -0.632]	4.92e-05	89/137

5 Synthetic corruptions and robustness

Table 5 Synthetic-corruption configuration in canonical units.

Corruption	Setting
Independent joint dropout	probability 0.05
Temporal block dropout	joint probability 0.25; 16 frames
Impulse noise	probability 0.02; scale 0.10
Random-walk drift	joint probability 0.10; step scale 0.01
Thorax rotation bias	frame probability 0.10; $\pm 12^\circ$
Frozen segment	joint probability 0.10; 12 frames
Integer time shift	joint probability 0.10; at most 4 frames

Table 6 Output displacement under the fixed corruption manifest (27 validation participants, one seed). Lower values indicate less movement from each method's own clean output, not recovery toward independent ground truth.

Challenge	A3	A6
Joint dropout	0.0629	0.0873
Temporal block dropout	0.0619	0.0873
Impulse noise	0.0605	0.0605
Random-walk drift	0.0445	0.0445
Thorax rotation bias	0.0312	0.0312
Frozen segment	0.0643	0.0643
Integer time shift	0.0348	0.0348

A6 had greater aggregate clean-to-corrupted displacement than A4 and A5 (0.120 versus approximately 0.09 repository units). A temporal-offset sweep over all 137 participants moved A6 by 0.046 and 0.078 units at offsets of ± 2 and ± 4 frames, respectively, essentially matching the deterministic mean. The learned model therefore did not resolve dependence on the upstream global offset.

6 Additional external comparisons

Two Unity-supervised comparators were evaluated over two direction-held-out folds and three seeds. An exact-extrinsic gate over the two 3D streams averaged 153.645 mm and 41.090°, a 6.74% improvement over its matched direct average. Learnable triangulation averaged 28.058 mm and 14.555°, 1.88% below its matched fixed-triangulation baseline. These rows use Unity native 3D during training and answer a different question from zero-shot private-model transfer.

The fitted-camera real-data pilot used the 14 held-out participants and three seeds. The frozen A6 baseline averaged 60.578 mm; the full camera-feature model averaged 60.850 mm, and its 30-degree wrong-camera control averaged 60.850 mm. Correct-camera features therefore failed the preregistered gate and are retained as a negative result rather than a positive comparator.

7 Complete per-joint comparison

The following descriptive table uses the same 14 held-out participants and the same sequence-similarity-plus-hip-centring evaluator for every method. It does not constitute 70 separate hypothesis tests.

Table 7: Complete MHR70 per-joint agreement on the same 14 held-out participants. Values are participant-level mean MPJPE in mm; bold is the lowest descriptive value in each row. Each cycle uses one similarity alignment followed by framewise hip centring.

Joint	Face	Side	Body mean	A6	Extrinsic-R	Extrinsic-R quality
nose	65.3	**38.8**	40.8	40.0	39.9	41.3
left-eye	65.5	41.7	40.6	**40.2**	40.5	42.1
right-eye	64.0	41.7	40.2	39.3	**39.2**	41.1
left-ear	61.5	40.0	38.2	**38.1**	39.1	40.5
right-ear	56.9	39.2	36.7	35.6	**35.4**	37.0
left-shoulder	43.7	31.1	31.3	**30.9**	31.2	31.7
right-shoulder	40.0	29.0	29.7	27.9	**27.0**	27.5
left-elbow	47.1	47.9	38.4	39.1	**36.8**	38.1
right-elbow	47.8	47.8	38.2	37.9	**35.4**	35.8
left-hip	6.4	8.7	6.1	6.3	**5.3**	5.4
right-hip	6.4	8.7	6.1	6.3	**5.3**	5.4
left-knee	27.5	40.9	27.7	28.3	**27.2**	27.7
right-knee	**21.6**	36.1	25.7	24.8	24.4	24.4
left-ankle	**42.5**	73.0	53.0	51.5	51.6	51.7
right-ankle	**37.0**	65.4	49.6	47.6	47.9	47.4
left-big-toe-tip	**47.1**	82.7	57.2	56.7	57.2	57.6
left-small-toe-tip	**45.1**	81.4	55.9	55.5	55.4	55.8
left-heel	**45.5**	77.5	57.2	55.3	55.6	55.6
right-big-toe-tip	**39.9**	73.0	52.1	50.1	50.6	50.4
right-small-toe-tip	**40.7**	75.0	54.0	51.8	52.2	52.0
right-heel	**40.7**	69.4	53.5	51.4	51.8	51.3
right-thumb-tip	109.1	104.8	82.0	80.4	80.7	**79.6**
right-thumb-first-joint	104.8	99.8	78.7	77.1	77.3	**76.3**
right-thumb-second-joint	99.4	93.3	74.3	72.9	72.8	**71.9**
right-thumb-third-joint	93.8	87.4	70.0	68.8	68.3	**67.6**
right-index-tip	112.5	111.3	85.8	84.3	84.5	**83.5**
right-index-first-joint	110.2	108.7	84.1	82.6	82.7	**81.7**
right-index-second-joint	107.4	105.4	81.8	80.5	80.4	**79.5**
right-index-third-joint	102.2	98.6	77.4	76.1	75.9	**75.1**
right-middle-tip	110.1	109.6	83.9	82.6	82.4	**81.5**
right-middle-first-joint	108.0	107.3	82.4	81.2	80.9	**80.1**
right-middle-second-joint	106.0	104.9	80.9	79.7	79.3	**78.6**
right-middle-third-joint	101.2	98.1	76.5	75.6	74.8	**74.2**
right-ring-tip	107.6	107.1	81.7	80.5	80.0	**79.2**
right-ring-first-joint	105.7	104.9	80.3	79.2	78.6	**77.9**
right-ring-second-joint	103.8	102.3	78.8	77.9	77.0	**76.5**
right-ring-third-joint	99.1	96.0	74.7	73.9	72.8	**72.4**

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Table 7 – continued

Joint	Face	Side	Body mean	A6	Extrinsic-R	Extrinsic-R quality
right-pinky-tip	102.3	102.9	77.9	77.0	76.0	**75.5**
right-pinky-first-joint	101.0	101.0	76.8	76.0	74.9	**74.4**
right-pinky-second-joint	99.7	98.7	75.6	75.0	73.6	**73.3**
right-pinky-third-joint	96.7	93.7	72.7	72.1	70.7	**70.4**
right-wrist	89.3	83.2	66.4	65.6	64.4	**64.1**
left-thumb-tip	102.0	92.7	75.9	77.8	**73.6**	75.2
left-thumb-first-joint	98.1	88.1	72.7	74.6	**70.5**	72.1
left-thumb-second-joint	93.2	82.4	68.7	70.5	**66.5**	68.0
left-thumb-third-joint	88.2	77.0	64.8	66.5	**62.5**	64.1
left-index-tip	105.9	96.5	79.2	81.1	**76.6**	78.2
left-index-first-joint	104.0	94.7	77.9	79.8	**75.2**	76.9
left-index-second-joint	101.7	92.4	76.1	78.0	**73.4**	75.1
left-index-third-joint	96.8	86.6	71.9	73.7	**69.3**	70.9
left-middle-tip	105.2	96.2	78.5	80.5	**75.9**	77.6
left-middle-first-joint	103.4	94.2	77.3	79.3	**74.6**	76.3
left-middle-second-joint	101.5	92.0	75.8	77.8	**73.1**	74.8
left-middle-third-joint	96.5	85.6	71.5	73.3	**68.8**	70.4
left-ring-tip	103.1	94.1	76.8	78.8	**74.2**	75.9
left-ring-first-joint	101.3	92.0	75.5	77.5	**72.9**	74.6
left-ring-second-joint	99.4	89.5	74.1	75.9	**71.4**	73.1
left-ring-third-joint	94.9	83.6	70.1	71.9	**67.5**	69.1
left-pinky-tip	99.5	90.3	74.0	75.9	**71.4**	73.2
left-pinky-first-joint	98.1	88.5	73.0	74.9	**70.4**	72.1
left-pinky-second-joint	96.5	86.4	71.8	73.6	**69.1**	70.8
left-pinky-third-joint	93.0	81.5	68.6	70.3	**66.0**	67.6
left-wrist	84.8	72.9	62.0	63.7	**59.6**	61.2
left-olecranon	47.8	50.4	39.5	40.2	**37.8**	39.2
right-olecranon	47.8	50.1	39.0	38.9	**36.2**	36.9
left-cubital-fossa	47.9	47.1	38.4	39.0	**36.8**	38.1
right-cubital-fossa	49.6	48.1	38.9	38.4	**36.2**	36.4
left-acromion	47.0	**31.8**	32.8	32.6	32.9	33.5
right-acromion	42.4	29.1	30.6	28.7	**28.0**	28.6
neck	42.7	**27.3**	28.8	27.8	28.2	28.9

8 Complete cohort outcomes

Table 8 Out-of-fold cohort and repeated-cycle analysis. Cohort effects are elderly-minus-student mixed-model coefficients at normalized cycle position 0.5. MAD effects are participant-level median differences.

Outcome	Cohort effect [95% CI]	p_{Holm}	MAD effect (p_{Holm})
Axial rotation ROM	0.2475 [−0.0242, 0.5192]	0.2966	−0.0016 (1.0000)
Log angular speed (P95)	0.3006 [0.1136, 0.4876]	0.0114	0.0314 (0.4459)
Peak rotation phase	−0.0013 [−0.0568, 0.0542]	1.0000	0.0000 (1.0000)
Trunk tilt (P95)	0.0042 [0.0002, 0.0082]	0.2514	−0.0002 (1.0000)
Wrist wrapping (P95)	0.0289 [−0.0626, 0.1204]	1.0000	−0.0073 (1.0000)
Log cycle duration	−0.0127 [−0.0256, 0.0002]	0.2677	0.0083 (1.0000)
Log dimensionless jerk	0.4233 [0.2336, 0.6130]	0.0001	0.0377 (0.0208)
Log whole-body repeatability	−0.0145 [−0.1055, 0.0765]	1.0000	−0.0003 (1.0000)

All eight cohort-by-cycle-position interactions had $p_{\mathrm{Holm}} = 1.000$. No axial-angle or angular-speed phase cluster survived correction across descriptor families. Secondary phase-localized differences occurred for trunk tilt and wrist wrapping, but are hypothesis-generating and not anatomical validation. Omitting outer-fold fixed effects produced similar coefficients for log angular speed (0.2972) and log jerk (0.4161).